

Yunjae Won

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EDUCATION

KAIST Graduate School of AI

Integrated M.S./Ph.D., Advisor: Minjoon Seo

Seoul, Republic of Korea

Fall 2024 – Present

KAIST

Bachelor of Science in Electrical Engineering, Minor in Computer Science

Daejeon, Republic of Korea

Feb. 2018 – Aug. 2024

GPA: 3.76/4.3

PUBLICATION

Context-Informed Grounding Supervision

June 2025

*Hyunji Lee, Seunghyun Yoon, **Yunjae Won**, Hanseok Oh, Geewook Kim, Trung Bui, Franck Dernoncourt, Elias Stengel-Eskin, Mohit Bansal, Minjoon Seo*

Preprint. Under Review.

[arXiv](#)

Differential Information Distribution: A Bayesian Perspective on Direct Preference Optimization

***Yunjae Won**, Hyunji Lee, Hyeonbin Hwang, Minjoon Seo*

May 2025

Preprint. Under Review.

[arXiv](#)

EXPERIENCE

Foundation Model Research Intern

NAVER Cloud HyperCLOVA Pre-Training Team

Seongnam, Republic of Korea

Aug. 2025 – Nov. 2025

- Developed an efficient, scalable reasoning data-filtering pipeline using a lightweight 0.6B reward model, improving KMMLU accuracy by +4.72% and eliminating the need for previous resource-heavy (>80B) LLM-based methods.
- Led a large-scale data-cleansing project across 450K expert Q&A samples, removing 99.87% of detected PII and enhancing response quality (doubling the average answer length).
- Achieved an +8.82% accuracy gain on the MMLU-Pro benchmark by augmenting Q&A datasets with additional options and applying strict quality filters to newly generated reasoning traces.

Machine Learning Engineer Intern

NAVER G Place AI Development

Seongnam, Republic of Korea

Mar. 2023 – Aug. 2023

- Developed an easy-to-use distributed hyper-parameter optimization tool for a Kubernetes-based environment.
- Trained a light-weight accurate Image Quality Assessment model, reducing the model's size by over 10x while improving accuracy by 56%; model has been in-service since July 2023 and enabled a shift from GPU to CPU-based serving.

Undergraduate Research Assistant

Data Intelligence Lab, KAIST School of Electrical Engineering

Daejeon, Republic of Korea

Jan. 2023 – Feb. 2023

- Investigated new methods for incorporating contrastive learning techniques for active learning.
- Measured the performance degradation of popular Active Learning methods under instance-dependent and class-dependent label noise on the MNIST dataset.

Machine Learning Engineer Intern

NCSoft Speech AI Lab

Seongnam, Republic of Korea

July 2022 – Aug. 2022

- Developed an in-the-wild audio signal preprocessing pipeline for training Singing Source Extraction Models.
- Increased training data set size twelve-fold and trained a new model that outperformed the previous state-of-the-art, improving the Signal-to-Distortion Ratio from 8.06 to 8.45.
- Awarded 'Excellent NCSoft Summer Internship Project' and offered a 6-month extension.

PROJECTS

Generating Harmful Responses from Aligned Models

Spring 2025

- Project for KAIST AI707 <Advanced Topics in Deep Reinforcement Learning>
- Derived a closed-form expression for the ideal distribution of rejected responses under the DPO framework, showing the possibility of generating harmful responses from aligned policies.

Orthogonal Gradient Descent: Learning from Preferences with Minimal Forgetting

Fall 2024

- Project for KAIST AI611 <Deep Reinforcement Learning>
- Proposed a method using projected gradient descent to learn from human preferences while minimizing the forgetting of previous knowledge.

Metabolic Reaction Prediction via Next Token Prediction

Fall 2024

- Project for KAIST AI607 <Graph Mining and Social Network Analysis>
- Formulated the metabolic reaction prediction task as a next token prediction problem, proposing a transformer architecture to predict the next reaction.

Korean Text Recognition Challenge | 1st Place out of 1,158 participants

Jan. 2023

- Developed a Korean text recognition model on a dataset of handwritings from Korean children for the Kyowon Group AI Challenge.
- Outperformed an OCR corporate team backed by state-of-the-art GPUs using only limited Google Colaboratory resources.

Agricultural Products' Price Change Forecasting Challenge | 3rd Place out of 705 participants

Sep. 2022

- Constructed an Extra-Trees Regressor based time-series forecaster for a competition hosted by the Korea Agro-Fisheries & Food Trade Corporation.
- Devised a new data-augmentation technique using a polynomial trend based pseudo-labeling process.

Korean Face Open Set Verification

Fall 2022

- Project for KAIST EE488 <Deep Learning for Computer Vision>
- Trained an EfficientNet-based embedding network using metric learning on a crowd-sourced Korean celebrity face image dataset.
- Achieved the best model performance among models trained without any additional training data, leading to an invitation from the professor to present the development process.

RESEARCH INTERESTS

- Preference Optimization, Reinforcement Learning from Human Feedback
- Continual Learning, Knowledge Distillation
- Optimization
- Data-Centric AI